U.S. Patent No. 8,046,721 B2

Unlocking a device by performing gestures on an unlock image

CHARTED AGAINST iPhone slide to unlock (iOS 1 to 9), Apple

Patent owner's own product. Patent expired in 2025.

PATENT RECORD — EXHIBIT A-2

ASSIGNEE	Apple Inc
PRIORITY	Dec. 23, 2005
ISSUED	Oct. 25, 2011
EXPIRED	Dec. 23, 2025
STATUS	Expired - Lifetime

From patents.google.com, copy of Nov. 3, 2026. 3 of 3 record checks passed.

4	● 2	◐ 2	○ 0	4/4	3	32/35	Nov. 3, 2026
CLAIM ELEMENTS	DOCUMENTED	PARTIAL	NOT FOUND	QUOTES VERBATIM ON PAGE	SOURCES QUOTED	CHECKS PASSED	RESEARCHED AND CHECKED

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[pre] A method of unlocking a hand-held electronic device, the device including a touch-sensitive display, the method comprising:	● DOCUMENTED > "To unlock the first iPhone, you touched the arrow button on the lock screen and dragged it over to the right."[1] The original iPhone is unlocked by touching an on-screen arrow and dragging it right, documenting a touch-display unlocking method.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated June 29, 2017, during the patent term. 1 more supporting source in Appendix B.
1[a] detecting a contact with the touch-sensitive display at a first predefined location corresponding to an unlock image;	● DOCUMENTED > "To unlock the first iPhone, you touched the arrow button on the lock screen and dragged it over to the right."[2] For the original arrow-slider implementation, the user begins by touching the displayed arrow button before dragging it right, supporting an initial graphical touch target. This evidence should not be generalized to every version in iOS 1–9: iOS 7 allowed unlocking swipes from anywhere on the lock screen.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated June 29, 2017, during the patent term. 1 more supporting source in Appendix B.

1. *The original iPhone in photos: Looking back on the iPhone that started it all*, macworld.com (June 29, 2017), https://www.macworld.com/article/230231/original-2007-iphone-photo-album.html (last visited Nov. 3, 2026).
2. *Id.*

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[b] continuously moving the unlock image on the touch-sensitive display in accordance with movement of the contact while continuous contact with the touch screen is maintained, wherein the unlock image is a graphical, interactive user-interface object with which a user interacts in order to unlock the device; and	◐ PARTIAL > "As soon as you power it up, you're presented with a software button with an arrow on it that sits at the bottom of the screen in a channel with the message 'Slide to Unlock' that animates to suggest that you slide the button to the right."[3] The review describes a graphical arrow button in a channel that users slide right, but does not explicitly describe uninterrupted contact or continuous tracking of the image with the finger. An iOS 7 review separately reports movement of a whole screen layer rather than a small slider, so the earlier mechanism is not established across the full iOS 1–9 range.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page 1 more supporting source in Appendix B.
1[c] unlocking the hand-held electronic device if the moving the unlock image on the touch-sensitive display results in movement of the unlock image from the first predefined location to a predefined unlock region on the touch-sensitive display.	◐ PARTIAL > "Either way, the on-screen slider appears, but your iPhone doesn't actually awaken until you drag the slider to the right with your finger."[4] The guide links a rightward finger drag of the on-screen slider to unlocking the iPhone. It does not identify the slider's required destination or establish that reaching a predefined display region is the condition that triggers unlocking.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated Oct. 6, 2021, during the patent term.

3. *iPhone 1.0: The 10th Anniversary MacStories Review*, macstories.net, https://www.macstories.net/reviews/iphone-1-0-the-10th-anniversary-macstories-review (last visited Nov. 3, 2026).
4. *How to Lock and Unlock Your iPhone*, dummies.com (Oct. 6, 2021), https://www.dummies.com/article/technology/electronics/cell-phones/iphones/how-to-lock-and-unlock-your-iphone-196930 (last visited Nov. 3, 2026).

Appendix A · Prior art

Qualifying references need source metadata supporting a date before Dec. 23, 2005. Other candidates and the reasons for exclusion appear below.

No non-patent prior art passed the checks.

EXCLUDED OR AWAITING DATE VERIFICATION

1	*Touchscreen Toggle Switches: Push or Slide? Design Issues and Usability Study* http://www.cs.umd.edu/hcil/trs/90-08/90-08.pdf	● Published before the priority date · research: 1990-11; page metadata: unavailable; priority: 2005-12-23
2	*N1 Quick Start Guide, V0.5* https://www.courtlistener.com/docket/4178994/2121/15/apple-inc-v-samsung-electronics-co-ltd	● Published before the priority date · research: Not printed; publicly archived on 2004-07-29, before the priority date; page metadata: unavailable; priority: 2005-12-23

Appendix B · Research basis

For each element: Parallel's reasoning, its confidence, and every source it cited. The quoted source is marked ▸.

1[pre] DOCUMENTED · CONFIDENCE MEDIUM

Macworld directly describes touching and dragging an on-screen arrow to unlock the original iPhone, and its title/date segments establish the source metadata. MacStories explicitly identifies the iPhone's touch screen. These are review sources rather than primary technical documentation.

▸ Original iPhone 2007 Photo Album | Macworld · www.macworld.com/article/230231/original-2007-iphone-photo-album.html
iPhone 1.0: The 10th Anniversary MacStories Review - MacStories · www.macstories.net/reviews/iphone-1-0-the-10th-anniversary-macstories…

1[a] DOCUMENTED · CONFIDENCE MEDIUM

The Macworld sentence identifies the displayed arrow as the initial touch target for the original implementation. The iOS 7 description permits swiping anywhere on the lock screen, supporting the explicit limitation on generalizing that fixed-target evidence across the entire requested version range.

▸ Original iPhone 2007 Photo Album | Macworld · www.macworld.com/article/230231/original-2007-iphone-photo-album.html
Get to know iOS 7: Design changes | Macworld · www.macworld.com/article/221891/get-to-know-ios-7-design-changes.html

1[b] PARTIAL · CONFIDENCE MEDIUM

MacStories describes a software arrow button in a channel and direct finger manipulation of screen elements, supporting the graphical interactive-object portion. It does not explicitly describe uninterrupted contact or continuous image tracking. AppleInsider directly reports that iOS 7 moves an entire screen layer instead of the small slider, supporting the version caveat; no publication date was shown for the quoted MacStories page.

▸ iPhone 1.0: The 10th Anniversary MacStories Review - MacStories · www.macstories.net/reviews/iphone-1-0-the-10th-anniversary-macstories…
Inside Apple's iOS 7: Details on the new features of your just-updated iPhone & iPad | AppleInsider · appleinsider.com/articles/13/09/18/inside-apples-ios-7-details-on-the…

1[c] PARTIAL · CONFIDENCE MEDIUM

The guide's unlocking context and quoted instruction connect rightward finger dragging of the slider with unlocking. The cited passage supplies no required endpoint or predefined destination region, so it supports only a partial assessment. The page explicitly supplies the title and update date.

▸ How to Lock and Unlock Your iPhone | dummies · www.dummies.com/article/technology/electronics/cell-phones/iphones/ho…

Appendix C · Verification log

Every check run on this exhibit. Green passed, amber needs a person to look, red failed; details for every flag follow the grid.

CLAIM LANGUAGE	EVIDENCE	PRIOR ART	QUOTES, SOURCES AND DATES	PREPARED WITH
Extract API on patents.google.com, copy of Nov. 3, 2026. Split into elements in Python; no model writes claim text.	Task API, pro processor, patent databases excluded. completed	Task API, pro processor, dates enforced in Python. completed	Extract requests fresh copies. Python checks for a continuous quote and compares prior-art dates with publication metadata.	parallel

PATENT RECORD	PATENT NUMBER	CLAIM 1 VERBATIM	DATES IN ORDER
US8046721B2	●	●	●

CLAIM ELEMENT	RESEARCHED	QUOTE AND SOURCE	QUOTE ON PAGE	NON-PATENT SOURCE	IN RESEARCH BASIS	DATED
1[pre]	●	●	●	●	●	●
1[a]	●	●	●	●	●	●
1[b]	●	●	●	●	●	●
1[c]	●	●	●	●	●	●

PRIOR ART	BEFORE PRIORITY DATE	NON-PATENT SOURCE	QUOTE ON PAGE	LABELS MATCH CLAIM
Ref. 1	●	●	●	●
Ref. 2	●	●	●	●

FLAGGED FOR REVIEW · 3

- 1[b] · Source dated · no date on the page
- Ref. 1 · Published before the priority date · research: 1990-11; page metadata: unavailable; priority: 2005-12-23
- Ref. 2 · Published before the priority date · research: Not printed; publicly archived on 2004-07-29, before the priority date; page metadata: unavailable; priority: 2005-12-23